Lecture 1

Lie algebras, ideals and representations

Objectives. *Matrix commutators lead to the definition of a Lie algebra. We study ideals, quotients and representations through three small examples. Unlike complex representations of finite groups, modules of a Lie algebra need not split into irreducible submodules.*

By the end of this lecture you will be able to:

- Check the Lie algebra axioms in a matrix example.
- Find the centre, derived algebra and quotient of a small Lie algebra.
- Construct a representation and identify its invariant subspaces.
- Prove Schur's lemma for finite-dimensional irreducible complex modules.

Throughout this lecture, vector spaces and Lie algebras are finite-dimensional over $\mathbb{C}$. Products of linear maps mean composition. The product AB applies B first.

1.1 The bracket

For two endomorphisms A and B of a vector space V, define their *commutator* by

$$[A, B] = AB - BA.$$

The commutator measures the failure of A and B to commute. It is bilinear and satisfies $[A, B] = -[B, A]$.

Associativity of composition gives another identity. Expand the three double commutators:

$$\begin{aligned} [A, [B, C]] &= ABC - ACB - BCA + CBA, \\ [B, [C, A]] &= BCA - BAC - CAB + ACB, \\ [C, [A, B]] &= CAB - CBA - ABC + BAC. \end{aligned}$$

Each of the six products occurs once with each sign. Hence

$$[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0.$$

This is the *Jacobi identity*.

Definition 1.1 (Lie algebra)**.** A *Lie algebra* is a complex vector space $\mathfrak{g}$ with a bilinear map

$$[\,\cdot\,,\,\cdot\,]\colon \mathfrak{g}\times\mathfrak{g}\longrightarrow\mathfrak{g}$$

such that

$$[x,y]=-[y,x],\qquad [x,[y,z]]+[y,[z,x]]+[z,[x,y]]=0\quad\text{for all } x,y,z\in\mathfrak{g}.$$

The map is the *Lie bracket*.

Over $\mathbb{C}$, antisymmetry gives $2[x,x]=0$, so $[x,x]=0$. Conversely, bilinearity and $[x,x]=0$ imply antisymmetry by expanding $[x+y,x+y]$.

The vector space $\mathrm{End}(V)$ with the commutator bracket is denoted $\mathfrak{gl}(V)$. A basis of V identifies $\mathfrak{gl}(V)$ with all complex $n\times n$ matrices, where $n=\dim V$. We denote this Lie algebra by $\mathfrak{gl}_n(\mathbb{C})$.

Definition 1.2 (Subalgebra and abelian algebra)**.** A vector subspace $\mathfrak{a}\subseteq\mathfrak{g}$ is a *Lie subalgebra* if

$$[x,y]\in\mathfrak{a}\quad\text{for all } x,y\in\mathfrak{a}.$$

A Lie algebra is *abelian* if every bracket is zero.

Every vector space becomes an abelian Lie algebra by giving it the zero bracket.

For $A,B\in\mathfrak{gl}_n(\mathbb{C})$, the trace satisfies

$$\operatorname{tr}(AB)=\sum_{i,j}A_{ij}B_{ji}=\sum_{j,i}B_{ji}A_{ij}=\operatorname{tr}(BA).$$

Every matrix commutator is therefore traceless. Hence

$$\mathfrak{sl}_n(\mathbb{C})=\{X\in\mathfrak{gl}_n(\mathbb{C}):\operatorname{tr}X=0\}$$

is a Lie subalgebra. It is the *special linear Lie algebra*.

Example 1.3 (The algebra $\mathfrak{sl}_2(\mathbb{C})$)**.** Use the ordered basis (h,e,f), where

$$h=\begin{pmatrix}1&0\\0&-1\end{pmatrix},\qquad e=\begin{pmatrix}0&1\\0&0\end{pmatrix},\qquad f=\begin{pmatrix}0&0\\1&0\end{pmatrix}.$$

Every traceless 2×2 matrix has the unique form

$$\begin{pmatrix}a&b\\c&-a\end{pmatrix}=ah+be+cf.$$

For example,

$$ef=\begin{pmatrix}1&0\\0&0\end{pmatrix},\qquad fe=\begin{pmatrix}0&0\\0&1\end{pmatrix}.$$

Hence $[e,f]=h$. The other two brackets are

$$[h,e]=2e,\qquad [h,f]=-2f.$$

These three relations determine every bracket by bilinearity and antisymmetry.

The product $h^2=I_2$ is not traceless. Thus $\mathfrak{sl}_2(\mathbb{C})$ is closed under the bracket, but not under matrix multiplication. We also write $\mathfrak{sl}_2$ for this algebra.

1.2 Ideals and quotients

For subspaces $U, W \subseteq \mathfrak{g}$, write

$$[U, W] = \operatorname{span}\{[u, w] : u \in U,\ w \in W\}.$$

This notation means the span of the brackets, not just the set of brackets.

Definition 1.4 (Ideal, centre and derived algebra)**.** An *ideal* of $\mathfrak{g}$ is a vector subspace $\mathfrak{a}$ such that

$$[\mathfrak{g}, \mathfrak{a}] \subseteq \mathfrak{a}.$$

The *centre* and the *derived algebra* of $\mathfrak{g}$ are

$$Z(\mathfrak{g}) = \{z \in \mathfrak{g} : [x, z] = 0 \text{ for all } x \in \mathfrak{g}\}, \qquad [\mathfrak{g}, \mathfrak{g}] = \operatorname{span}\{[x, y] : x, y \in \mathfrak{g}\}.$$

Every ideal is a subalgebra. The converse fails. For example, $\mathbb{C}h$ is a subalgebra of $\mathfrak{sl}_2$, but it is not an ideal:

$$[e, h] = -2e \notin \mathbb{C}h.$$

The centre is an ideal, because $[\mathfrak{g}, Z(\mathfrak{g})] = 0$. The derived algebra is also an ideal. For $x \in \mathfrak{g}$ and $w \in [\mathfrak{g}, \mathfrak{g}]$, the element $[x, w]$ is a bracket of two elements of $\mathfrak{g}$. Hence $[x, w] \in [\mathfrak{g}, \mathfrak{g}]$.

Proposition 1.5 (Quotient algebra)**.** *Let $\mathfrak{a}$ be an ideal of $\mathfrak{g}$. The quotient vector space $\mathfrak{g}/\mathfrak{a}$ has a Lie bracket given by*

$$[x + \mathfrak{a}, y + \mathfrak{a}] = [x, y] + \mathfrak{a}.$$

The quotient map $q\colon \mathfrak{g} \to \mathfrak{g}/\mathfrak{a}$, given by $q(x) = x + \mathfrak{a}$, preserves brackets.

Proof. First check that the bracket of cosets does not depend on representatives. For $a, b \in \mathfrak{a}$, bilinearity gives

$$[x + a, y + b] - [x, y] = [x, b] + [a, y] + [a, b].$$

Each term belongs to $\mathfrak{a}$, by the ideal condition and antisymmetry. The two brackets therefore define the same coset.

Bilinearity, antisymmetry and Jacobi pass to cosets. For example, the Jacobi sum in the quotient is the coset of the Jacobi sum in $\mathfrak{g}$, so it is zero. The defining formula gives

$$q([x, y]) = [q(x), q(y)].$$ ∎

The ideal condition is necessary. If the displayed bracket on cosets is well defined, take $a \in \mathfrak{a}$. The cosets $a + \mathfrak{a}$ and $0 + \mathfrak{a}$ agree. Hence

$$[x, a] + \mathfrak{a} = [x, 0] + \mathfrak{a} = \mathfrak{a} \quad \text{for all } x \in \mathfrak{g},$$

which says $[x, a] \in \mathfrak{a}$.

Definition 1.6 (Homomorphism)**.** Let $\mathfrak{g}$ and $\mathfrak{k}$ be Lie algebras. A *Lie algebra homomorphism* is a linear map $\varphi\colon \mathfrak{g} \to \mathfrak{k}$ such that

$$\varphi([x, y]) = [\varphi(x), \varphi(y)] \quad \text{for all } x, y \in \mathfrak{g}.$$

A bijective homomorphism is an *isomorphism*.

The kernel of φ is an ideal. Indeed, if $a \in \ker \varphi$ and $x \in \mathfrak{g}$, then

$$\varphi([x, a]) = [\varphi(x), \varphi(a)] = [\varphi(x), 0] = 0.$$

The image is a subalgebra, since $[\varphi(x), \varphi(y)] = \varphi([x, y])$. The first isomorphism theorem for vector spaces gives a linear bijection

$$\mathfrak{g}/\ker\varphi \longrightarrow \operatorname{im}\varphi, \qquad x + \ker\varphi \longmapsto \varphi(x).$$

The map preserves brackets by the homomorphism identity.

The quotient $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$ is abelian. More generally, $\mathfrak{g}/\mathfrak{a}$ is abelian exactly when

$$[\mathfrak{g}, \mathfrak{g}] \subseteq \mathfrak{a}.$$

This follows directly from the quotient bracket. Thus $[\mathfrak{g}, \mathfrak{g}]$ is the smallest ideal with abelian quotient.

1.3 The adjoint representation

A Lie algebra acts on its own vector space through the bracket. For $x \in \mathfrak{g}$, define

$$\operatorname{ad} x \colon \mathfrak{g} \longrightarrow \mathfrak{g}, \qquad (\operatorname{ad} x)(y) = [x, y].$$

Rewrite Jacobi as

$$[x, [y, z]] = [[x, y], z] + [y, [x, z]]. \tag{1.1}$$

For $z \in \mathfrak{g}$, this gives

$$[\operatorname{ad} x, \operatorname{ad} y](z) = [x, [y, z]] - [y, [x, z]] = [[x, y], z] = \operatorname{ad}([x, y])(z).$$

Hence

$$\operatorname{ad} \colon \mathfrak{g} \longrightarrow \mathfrak{gl}(\mathfrak{g})$$

is a Lie algebra homomorphism. It is the *adjoint representation*. Since $\operatorname{ad} x = 0$ means $[x, y] = 0$ for all $y \in \mathfrak{g}$, its kernel is $Z(\mathfrak{g})$.

An ideal is exactly a subspace invariant under every operator $\operatorname{ad} x$. Thus the ideals are the invariant subspaces of the adjoint representation.

Definition 1.7 (Derivation)**.** A *derivation* of $\mathfrak{g}$ is a linear map $D \colon \mathfrak{g} \to \mathfrak{g}$ such that

$$D([y, z]) = [Dy, z] + [y, Dz] \quad \text{for all } y, z \in \mathfrak{g}.$$

A derivation of the form $\operatorname{ad} x$ is an *inner derivation*.

Equation (1.1) says that every $\operatorname{ad} x$ is a derivation. This is the bracket version of the product rule.

In a nonzero abelian algebra, both sides of the derivation rule vanish. Thus every linear map is a derivation, while every inner derivation is zero.

Example 1.8 (Centre and derived algebra of $\mathfrak{sl}_2$)**.** The bracket relations in Example 1.3 give

$$e = \frac{1}{2}[h, e], \qquad f = -\frac{1}{2}[h, f], \qquad h = [e, f].$$

Every basis vector is a bracket, so $[\mathfrak{sl}_2, \mathfrak{sl}_2] = \mathfrak{sl}_2$. To find the centre, let $x = ah + be + cf$. If x is central, then

$$[h, x] = 2be - 2cf = 0.$$

Independence of e and f gives $b = c = 0$. The remaining condition is

$$[e, ah] = -2ae = 0,$$

so $a = 0$. Hence $Z(\mathfrak{sl}_2) = 0$, and the adjoint representation is injective.

1.4 Representations and invariant subspaces

Definition 1.9 (Representation). A *representation* of $\mathfrak{g}$ on V is a Lie algebra homomorphism

$$\rho : \mathfrak{g} \longrightarrow \mathfrak{gl}(V).$$

Equivalently, ρ is linear and

$$\rho([x,y]) = \rho(x)\rho(y) - \rho(y)\rho(x) \quad \text{for all } x, y \in \mathfrak{g}.$$

The space V with this action is a $\mathfrak{g}$-*module*. The representation is *faithful* if $\ker \rho = 0$.

A group representation turns products into products of operators. A Lie algebra representation turns brackets into commutators of operators. The operators $\rho(x)$ of a Lie algebra representation need not be invertible.

If a basis $(v_1, \dots, v_d)$ of V is fixed, write $D(x)$ for the matrix of $\rho(x)$ in this basis. Its entries satisfy

$$\rho(x)v_j = \sum_{i=1}^{d} D(x)_{ij} v_i.$$

The *defining representation* of a matrix Lie algebra $\mathfrak{g} \subseteq \mathfrak{gl}_n(\mathbb{C})$ is its action on column vectors:

$$\rho_{\mathrm{def}}(X)v = Xv.$$

Another example is the *trivial representation*, where $\rho(x) = 0$ for every x. Unlike a trivial group representation, its operators are zero, not identity operators.

Definition 1.10 (Submodule and irreducibility). A subspace $W \subseteq V$ is a *submodule* if

$$\rho(x)W \subseteq W \quad \text{for all } x \in \mathfrak{g}.$$

A nonzero module is *irreducible* if its only submodules are 0 and V. A module is *completely reducible* if it is a direct sum of irreducible submodules.

The restricted action makes W a module. The quotient V/W is also a module, with

$$x \cdot (v + W) = \rho(x)v + W.$$

This is well defined because W is invariant. Here the acting algebra stays $\mathfrak{g}$. By contrast, the quotient in Proposition 1.5 changed the algebra itself.

For the defining $\mathfrak{sl}_2$-module, let (v_1, v_2) be the standard basis of $\mathbb{C}^2$. Suppose a submodule W contains $u = av_1 + bv_2 \neq 0$. If $b \neq 0$, then

$$e(av_1 + bv_2) = bv_1 \in W, \qquad f(bv_1) = bv_2 \in W.$$

Thus $W = \mathbb{C}^2$. If $b = 0$, then $a \neq 0$. The vector $fu = av_2$ reduces to the first case. Hence the defining module is irreducible.

Definition 1.11 (Intertwiner). For $\mathfrak{g}$-modules V and W, with representations ρ_V and ρ_W, an *intertwiner* is a linear map $T : V \to W$ such that

$$T\rho_V(x) = \rho_W(x)T \quad \text{for all } x \in \mathfrak{g}.$$

The vector space of intertwiners is denoted $\mathrm{Hom}_{\mathfrak{g}}(V, W)$. Write $\mathrm{End}_{\mathfrak{g}}(V) = \mathrm{Hom}_{\mathfrak{g}}(V, V)$. An invertible intertwiner is an isomorphism of modules.

The kernel and image of an intertwiner are submodules. For the kernel,

$$Tv = 0 \quad \Longrightarrow \quad T\rho_V(x)v = \rho_W(x)Tv = 0.$$

For the image, use

$$\rho_W(x)(Tv) = T(\rho_V(x)v).$$

Theorem 1.12 (Schur's lemma). *Let V and W be finite-dimensional irreducible complex $\mathfrak{g}$-modules. Every nonzero intertwiner $V \to W$ is an isomorphism. Moreover,*

$$\operatorname{End}_{\mathfrak{g}}(V) = \mathbb{C}\,\mathrm{id}_V\,.$$

Proof. The kernel of a nonzero intertwiner is not V, so it is zero. Its image is nonzero, so it is W. The map is therefore an isomorphism.

Now take $T \in \operatorname{End}_{\mathfrak{g}}(V)$. Since $\mathbb{C}$ is algebraically closed, T has an eigenvalue λ. The intertwiner $T - \lambda\,\mathrm{id}_V$ has a nonzero kernel. The first assertion forces this intertwiner to be zero. Hence $T = \lambda\,\mathrm{id}_V$. ∎

Schur's lemma does not say that every module is completely reducible. For example, let the one-dimensional abelian algebra $\mathbb{C}t$ act on $\mathbb{C}^2$. Choose $\rho(t)$ with matrix in the standard basis

$$D(t) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

The line $\mathbb{C}(1,0)^{\mathsf{T}}$ is invariant. Every complementary line has a spanning vector $(a,1)^{\mathsf{T}}$ for some $a \in \mathbb{C}$. But

$$D(t)\begin{pmatrix} a \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \notin \mathbb{C}\begin{pmatrix} a \\ 1 \end{pmatrix}.$$

Hence this is the only invariant line. The module is reducible, but it cannot be a direct sum of two invariant lines. Thus it is not completely reducible. Lecture 5 will prove complete reducibility under a condition on $\mathfrak{g}$.

1.5 Two algebras with abelian quotients

For each algebra, we compute brackets of basis vectors. We then use these brackets to find the centre.

Example 1.13 (Upper triangular matrices). Let $\mathfrak{b}$ be the algebra of all upper triangular 2×2 matrices. For

$$X = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}, \qquad Y = \begin{pmatrix} u & v \\ 0 & w \end{pmatrix},$$

matrix multiplication gives

$$[X,Y] = \begin{pmatrix} 0 & (a-d)v + (w-u)b \\ 0 & 0 \end{pmatrix}.$$

Every bracket lies in $\mathbb{C}e$. Since $[h,e] = 2e$, every multiple of e is a bracket. Therefore

$$[\mathfrak{b},\mathfrak{b}] = \mathbb{C}e.$$

A central X must commute with h and with e. The bracket formula gives

$$[X,h] = -2be, \qquad [X,e] = (a-d)e.$$

Hence $b = 0$ and $a = d$. Conversely, scalar matrices commute with every matrix. Thus

$$Z(\mathfrak{b}) = \mathbb{C}I_2.$$

Give $\mathbb{C}^2$ the zero bracket. The map

$$\mathfrak{b} \longrightarrow \mathbb{C}^2, \qquad \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \longmapsto (a, d)$$

is a surjective homomorphism with kernel $\mathbb{C}e$. Every bracket has zero diagonal, which verifies the homomorphism condition. Hence $\mathfrak{b}/\mathbb{C}e \cong \mathbb{C}^2$.

Example 1.14 (The Heisenberg algebra)**.** Inside $\mathfrak{gl}_3(\mathbb{C})$, take

$$p = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad q = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \qquad z = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Their span is the *Heisenberg algebra* $\mathfrak{h}_3$. The products give

$$[p, q] = z, \qquad [p, z] = [q, z] = 0.$$

Hence $[\mathfrak{h}_3, \mathfrak{h}_3] = \mathbb{C}z$.

For $x = ap + bq + cz$, the centre conditions are

$$[p, x] = bz = 0, \qquad [q, x] = -az = 0.$$

Thus $a = b = 0$, and

$$Z(\mathfrak{h}_3) = \mathbb{C}z.$$

The quotient by $\mathbb{C}z$ is abelian of dimension two.

Now let V be a nonzero $\mathfrak{h}_3$-module. Suppose $\rho(z) = c\ \mathrm{id}_V$. Then

$$[\rho(p), \rho(q)] = \rho([p, q]) = \rho(z) = c\ \mathrm{id}_V, \qquad 0 = \operatorname{tr}[\rho(p), \rho(q)] = c\ \dim V.$$

Over $\mathbb{C}$, this forces $c = 0$. Centrality of z gives

$$[\rho(z), \rho(x)] = \rho([z, x]) = 0 \quad \text{for all } x \in \mathfrak{h}_3.$$

Thus $\rho(z)$ is an intertwiner. In an irreducible module, Schur's lemma makes $\rho(z)$ scalar, so $\rho(z) = 0$.

The defining three-dimensional module is faithful. Its first coordinate line is invariant, since p, q and z all send $(1, 0, 0)^{\mathsf{T}}$ to zero. This module is reducible, and its nonzero operator $\rho(z)$ is not scalar.

Self-checks

(1) Is the span of h and e an ideal of $\mathfrak{sl}_2$?

(2) What is $\ker(\mathrm{ad}\colon \mathfrak{h}_3 \to \mathfrak{gl}(\mathfrak{h}_3))$?

(3) Can a nonzero finite-dimensional complex $\mathfrak{h}_3$-module V satisfy $[\rho(p), \rho(q)] = \mathrm{id}_V$?

Answers. (1) No. The bracket $[f, h] = 2f$ lies outside the span of h and e. (2) The kernel is the centre $\mathbb{C}z$. (3) No. Taking traces gives $0 = \dim V$.

Summary

A Lie bracket is bilinear and antisymmetric, and it satisfies the Jacobi identity. Ideals are exactly the subspaces for which the bracket of cosets is well defined. The centre is the kernel of the adjoint representation.

Representations turn brackets into commutators. Intertwiners between finite-dimensional irreducible complex modules satisfy Schur's lemma. A module need not be completely reducible. The next two lectures define nilpotency and solvability. They apply these definitions to the triangular examples.

Reading

Humphreys, *Introduction to Lie Algebras and Representation Theory*, Sections 1–2 and 6.1. Erdmann–Wildon, *Introduction to Lie Algebras*, Chapters 1–2 and 7. Exercise Sheet 1 gives further calculations and proofs.